Problem Set 4 Solutions
ECO3121 - Fall 2021

December 21, 2022

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Question 1

Solutions:

1. The model is given by:

$$Y_{it} = \beta_1 X_{it} + \alpha_i + u_{it}$$

The fact thatα_i is not observable poses an omitted variable problem in estimating our model by OLS. Let me then define $\Delta Y_{it} = Y_{it} - Y_{i,t-1}$, $\Delta X_{it} = X_{it} - X_{i,t-1}$ and $\Delta u_{it} = u_{it} - u_{i,t-1}$. Subtracting from the regression equation for Y_{it} its analogue at time $t-1$ we get:

$$\Delta Y_{it} = \beta_1 \Delta X_{it} + \Delta u_{it}$$

where fixed effects have been canceled out by taking time differences. At this point, we just need to estimate this model using OLS to get $\hat{\beta}_1$. A factor that can be modeled as a village fixed effect in this context can be given by the geographic location of the village itself, for example, rainfall, weather condition, etc.
Grading Rubric: +0.5 for estimation procedure, +0.5 for example of factor.

2. The model is now given by:

$$Y_{it} = \beta_1 X_{it} + \gamma_t + u_{it}$$

Again, the fact that γ_t is not observable, poses an omitted variable problem in estimating our model by OLS. Let me then define:

$$\overline{Y}_t = \frac{1}{N} \sum_{i=1}^{N} Y_{it} = \beta_1 \overline{X}_t + \gamma_t + \overline{u}_t$$

where X_t and u_t are the analogues of Y_t for X_{it} and u_{it}. Let's now consider the demeaned version of the model above:

$$Y_{it} - \overline{Y}_t = \beta_1 (X_{it} - \overline{X}_t) + u_{it} - \overline{u}_t$$

Let $\tilde{Y}_{it} = Y_{it} - \overline{Y}_t$ and, in the same fashion, construct $\tilde{X}_{it}$ and $\tilde{u}_{it}$ to rewrite:

$$\tilde{Y}_{it} = \beta_1 \tilde{X}_{it} + \tilde{u}_{it}$$

As before, we just need to estimate this model using OLS to get $\hat{\beta}_1$. A factor that can be modeled as a time fixed effect in this context can be given national agricultural tax policies. Clearly, this factor would affect different villages in the same way, and it's likely to change over time.
Grading Rubric: +0.5 for estimation procedure, +0.5 for example of factor.

3. The advantage of a double fixed effect model is that it takes into account both the village-fixed unobserved variables and the time-fixed unobserved variables, which minimizes the omitted variable bias risk. The estimation just repeat the above procedure with demean at the first village level, and then at the time level.

Question 2

Solutions:

1. - The differences-in-difference estimator for the causal effect of poverty alleviation program corresponds to β_1 in the regression,

$$\Delta Y_i = \beta_0 + \beta_1 D_i + u_i$$

for $Y_i = Y_i^1 - Y_i^0$.
Alternatively, we can write the treatment effect by $(\bar{Y}^{treated,1} - \bar{Y}^{treated,0}) - (\bar{Y}^{control,1} - \bar{Y}^{control,0})$, where $\bar{Y}^{treated,1}$ is the sample average of Y_i^1 among $D_i = 1$, i.e., farmers in Gansu province, and $\bar{Y}^{control,1}$ is the sample average of Y_i^1 among $D_i = 0$, i.e., neighborhoods without a Jewish intuition nearby, and the other two terms are defined similarly.

2. - The key assumption is the **parallel** trend assumption: without the program, the crop yield change $Y_i = Y_i^1 - Y_i^0$ on average would have been the same for the treatment group and the control group.

3. - Over estimate
   - The reason being that an increase in the effort should, all else equal, increase crop yield - this being correlated with fertilizer usage, β_1 would capture said effect.
   - Alternatively, we can explain this through violation of the parallel trend assumption. In this case, the crop yield in the treatment group would have gone up even without the treatment. Attributing this change to the effect of the treatment will result in an over estimation of the effect of the poverty alleviation program.

Question 3

1. It is confounded by the unobserved differences between NJ vs. PA. (Give a specific example or draw a graph to illustrate to get full credit)

2. The regression model is specified as follows:

$$Empl = \beta_0 + \beta_1 Treatment + \beta_2 Group + \beta_3 Treatment * Group + \alpha_1 X + u$$

   where X stands for the series of control variables.

3. Key assumption: parallel trend assumption, i.e. without the treatment, the treated group and the control group were to move in the parallel trend over time.

4. Stata command:
   reg Empl i.Treatment i.Group i.Treatment # i.Group X
   (X denotes the series of control variables)
   or
   *gen Treatment_Group=Treatment * Group*
   reg Empl Treatment Group TreatmentGroup X

5. Draw the graph and the 2*2 table as in the slides, and show that the interaction term estimate the "true" treatment effect compared.